# AI Assisted Coding Ass-9.5

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## Problem 1: String Utilities Function

**Consider the following Python**

**function: def**

**reverse_string(text):**

**return text[::-1]**

## Task:

1. **Write documentation in:**
   - **(a) Docstring**
   - **(b) Inline comments**
   - **(c) Google-style documentation**
2. **Compare the three documentation styles.**

**Recommend the most suitable style for a utility-based string library.**

**Prompt: # Generate a Docstring Inline comments Google-style documentation to the reversing a string function.**

**Code:**

```python
def reverse_string(s):
    """
    Reverses the given string.

    Args:
        s (str): The string to be reversed.

    Returns:
        str: The reversed string.
    """
    # Initialize an empty string to store the reversed result
    reversed_str = ""

    # Iterate through the input string in reverse order
    for char in s:
        reversed_str = char + reversed_str  # Prepend the current character to the result
```

Output:

```
PS D:\3-2\AI Assitant Coding\Code Files> python -m pydoc Task95
Original string: Hello, World!
Reversed string: !dlroW ,olleH
Help on module Task95:

NAME
    Task95 - # Generate a Docstring Inline comments Google-style documentation to the reversing a string function.

FUNCTIONS
    reverse_string(s)
        Reverses the given string.

        Args:
            s (str): The string to be reversed.

        Returns:
            str: The reversed string.
```

Pydoc: Index of Modules | Python: module Task95

File D:/3-2/AI%20Assitant%20Coding/Code%20Files/Task95.html

index
**Task95** d:\3-2\ai assitant coding\code files\task95.py

# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.

**Functions**

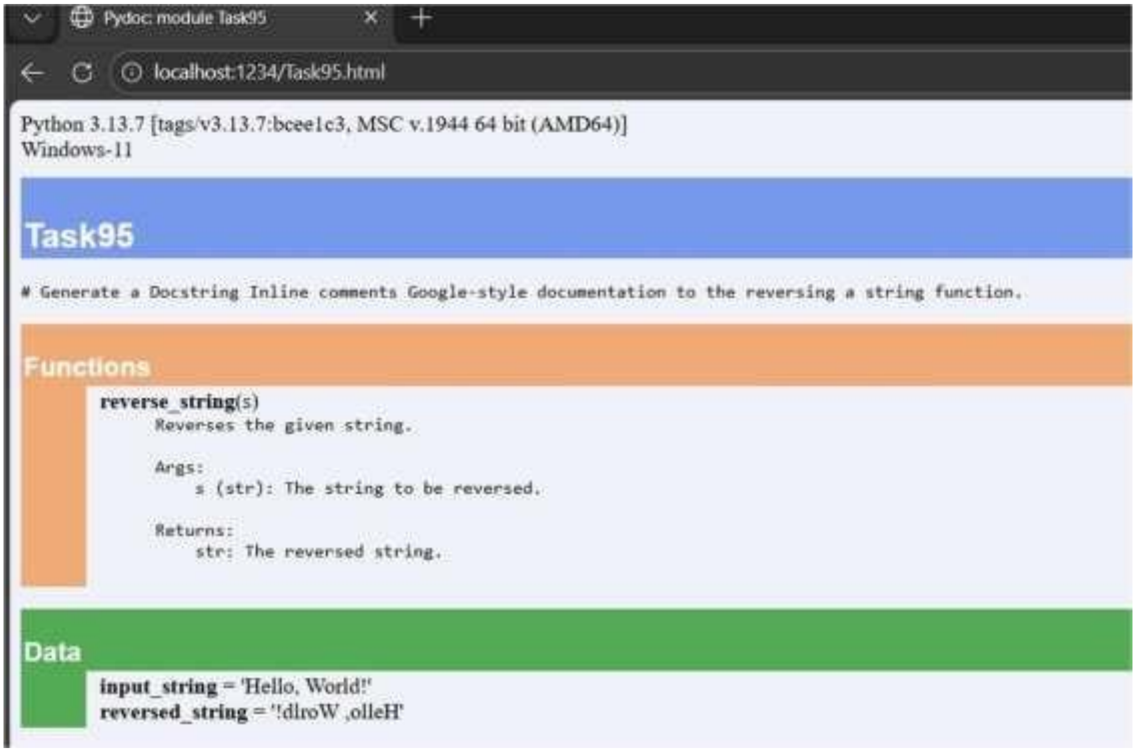
**reverse_string**(s)
Reverses the given string.

Args:
s (str): The string to be reversed.

Returns:
str: The reversed string.

**Data**

**input_string** = 'Hello, World!'
**reversed_string** = '!dlroW ,olleH'




# Problem 2: Password Strength Checker

**Consider the function:**

**def**

**check_strength(password**

**): return len(password)**

**>= 8**

## Task:

1. **Document the function using docstring, inline comments, and Google style.**
2. **Compare documentation styles for security-related code.**
3. **Recommend the most appropriate style.**

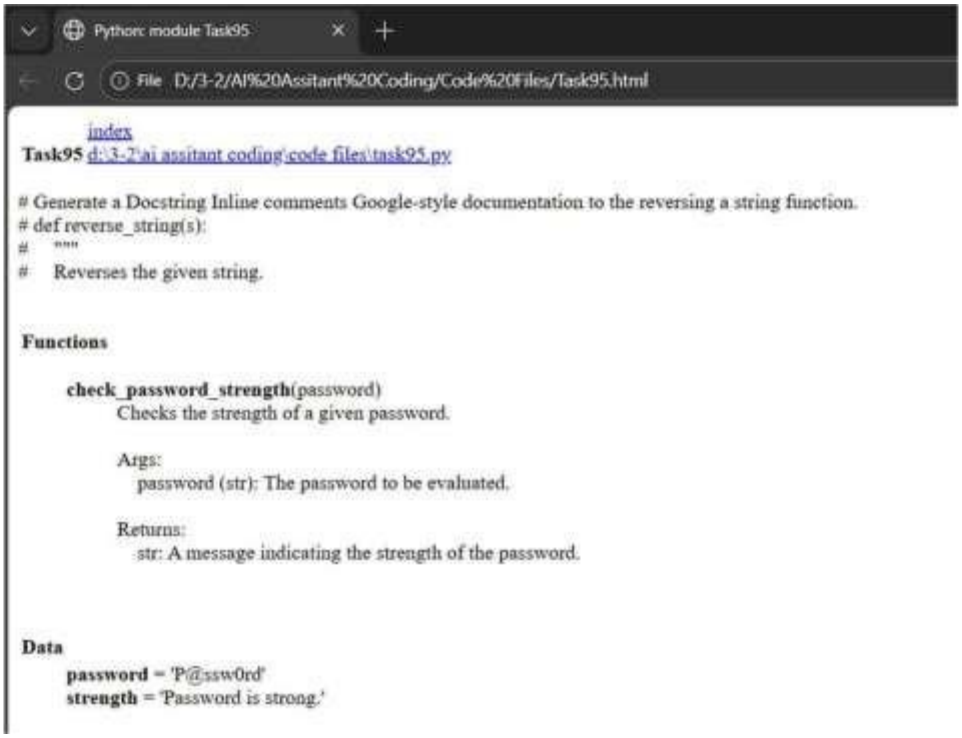
**Prompt: generate a 1. Document the function using docstring, inline comments, and Google style 2. Compare documentation styles for security-related code. for Password Strength Checker function**

**Code:**

```python
def check_password_strength(password):
    """
    Checks the strength of a given password.

    Args:
        password (str): The password to be evaluated.

    Returns:
        str: A message indicating the strength of the password.
    """
    # Initialize variables to track the presence of different character types
    has_upper = False
    has_lower = False
    has_digit = False
    has_special = False

    # Check the length of the password
    if len(password) < 8:
        return "Password is too short. It should be at least 8 characters long."

    # Iterate through each character in the password to check for different types
    for char in password:
        if char.isupper():
            has_upper = True  # Found an uppercase letter
        elif char.islower():
```

**Output:**

```
PS D:\3-2\AI Assitant Coding\Code Files> python -m pydoc Task95
Password: P@ssw0rd
Strength: Password is strong.
Help on module Task95:

NAME
    Task95

DESCRIPTION
    # Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
    # def reverse_string(s):
    #     """
    #     Reverses the given string.

FUNCTIONS
    check_password_strength(password)
        Checks the strength of a given password.

        Args:
            password (str): The password to be evaluated.

        Returns:
            str: A message indicating the strength of the password.
```




index
**Task95** d:\3-2\ai assitant coding\code files\task95.py

# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
# def reverse_string(s):
#     """
#     Reverses the given string.

**Functions**

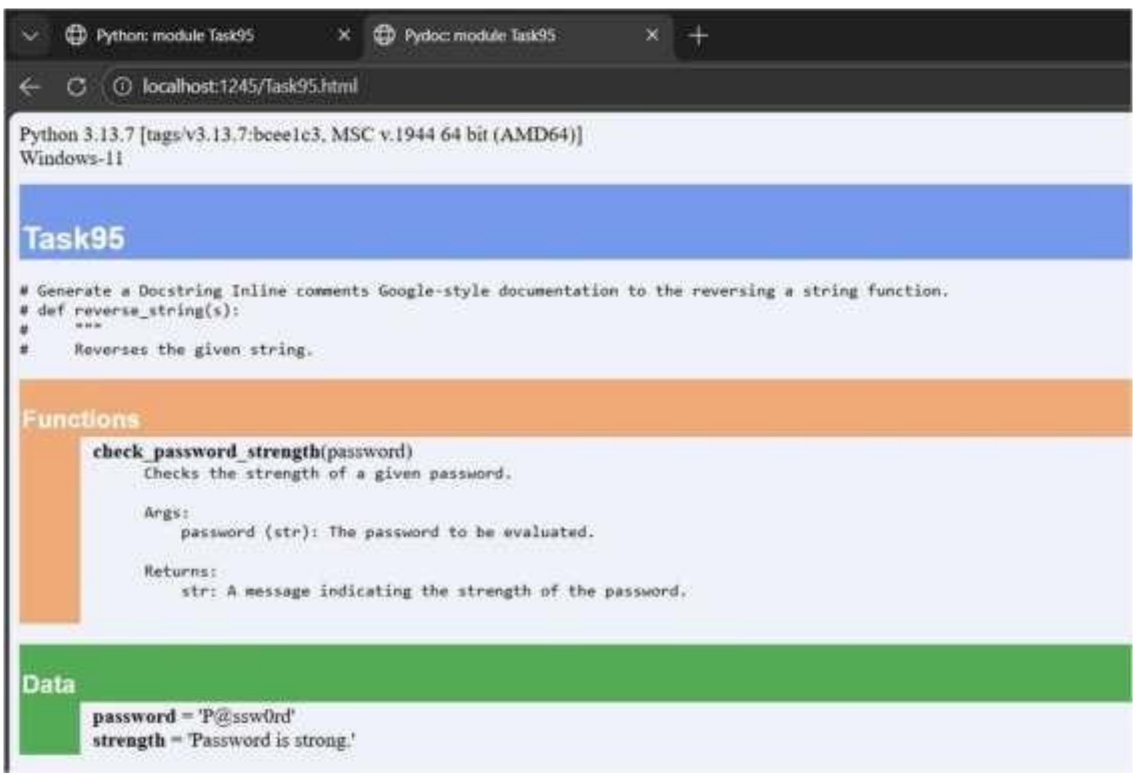
**check_password_strength**(password)
Checks the strength of a given password.

Args:
password (str): The password to be evaluated.

Returns:
str: A message indicating the strength of the password.

**Data**

**password** = 'P@ssw0rd'
**strength** = 'Password is strong.'




Python 3.13.7 [tags/v3.13.7:bcee1c3, MSC v.1944 64 bit (AMD64)]
Windows-11

**Task95**

```
# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
# def reverse_string(s):
#     """
#     Reverses the given string.
```

**Functions**

**check_password_strength**(password)

```
Checks the strength of a given password.

Args:
    password (str): The password to be evaluated.

Returns:
    str: A message indicating the strength of the password.
```

**Data**

**password** = 'P@ssw0rd'
**strength** = 'Password is strong.'

# Problem 3: Math Utilities Module

# Task:

1. **Create a module math_utils.py with functions:**
   - **square(n)**
   - **cube(n)**
   - **factorial(n)**
2. **Generate docstrings automatically using AI tools.**
3. **Export documentation as an HTML file.**

**Prompt: # Generate a 1. Create a module math_utils.py with functions:,square(n),cube(n),factorial(n),**
**# Generate docstrings automatically Code:**

```
def square(n):
    """
    Returns the square of a number.

    Args:
        n (int or float): The number to be squared.

    Returns:
        int or float: The square of the input number.
    """
    return n * n

def cube(n):
    """
    Returns the cube of a number.

    Args:
        n (int or float): The number to be cubed.

    Returns:
        int or float: The cube of the input number.
    """
    return n * n * n

def factorial(n):
```

**Output:**

```
PS D:\3-2\AI Assitant Coding\Code Files> python -m pydoc Task95
Help on module Task95:

NAME
    Task95

DESCRIPTION
    # Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
    # def reverse_string(s):
    #     """
    #     Reverses the given string.

FUNCTIONS
    cube(n)
        Returns the cube of a number.

        Args:
            n (int or float): The number to be cubed.

        Returns:
            int or float: The cube of the input number.

    factorial(n)
        Returns the factorial of a number.
```

Python: module Task95

File D:/3-2/AI%20Assitant%20Coding/Code%20Files/Task95.html

index
**Task95** d:\3-2\ai assitant coding\code files\task95.py

\# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
\# def reverse_string(s):
\#     """
\#     Reverses the given string.

**Functions**

**cube**(n)
Returns the cube of a number.

Args:
n (int or float): The number to be cubed.

Returns:
int or float: The cube of the input number.

**factorial**(n)
Returns the factorial of a number.

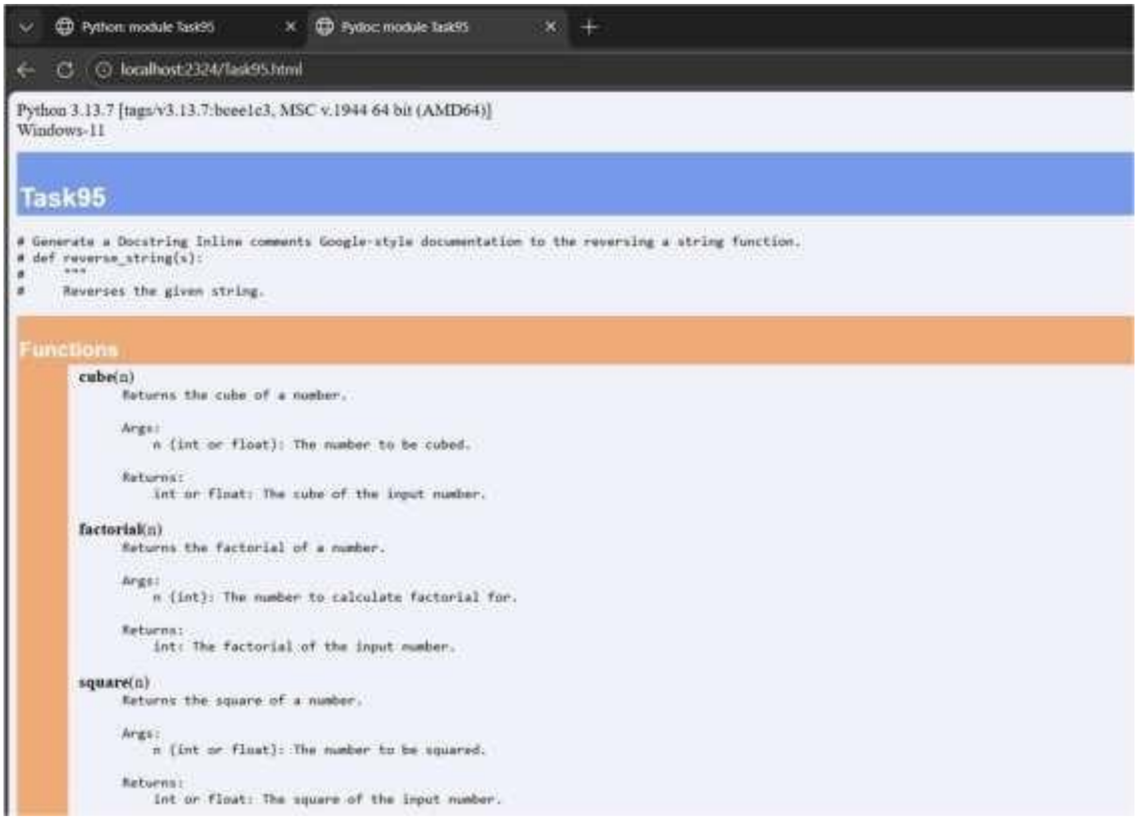
Args:
n (int): The number to calculate factorial for.

Returns:
int: The factorial of the input number.

**square**(n)
Returns the square of a number.

Args:
n (int or float): The number to be squared.

Returns:
int or float: The square of the input number.



Python 3.13.7 [tags/v3.13.7:bcee1c3, MSC v.1944 64 bit (AMD64)]
Windows-11

Task95

```
# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
# def reverse_string(s):
#     """
#     Reverses the given string.
```

Functions

**cube**(n)
```
    Returns the cube of a number.

    Args:
        n (int or float): The number to be cubed.

    Returns:
        int or float: The cube of the input number.
```

**factorial**(n)
```
    Returns the factorial of a number.

    Args:
        n (int): The number to calculate factorial for.

    Returns:
        int: The factorial of the input number.
```

**square**(n)
```
    Returns the square of a number.

    Args:
        n (int or float): The number to be squared.

    Returns:
        int or float: The square of the input number.
```

# Problem 4: Attendance Management Module

# Task:

1. **Create a module attendance.py with functions:**
    - **mark_present(student)**
    - **mark_absent(student)**
    - **get_attendance(student)**
2. **Add proper docstrings.**
3. **Generate and view documentation in terminal**

**and browse Prompt: Generate 1.Create a module attendance.py with functions:mark_present(student),mark_absent(student),**

**# get_attendance(student) 3.Generate and view documentation**

**Code:**

```python
class Attendance:
    def __init__(self):
        """
        Initializes the Attendance class with an empty attendance record.
        """
        self.attendance_record = {}

    def mark_present(self, student):
        """
        Marks a student as present.

        Args:
            student (str): The name of the student to be marked present.
        """
        self.attendance_record[student] = "Present"

    def mark_absent(self, student):
        """
        Marks a student as absent.

        Args:
```

**Output:**

```
PS D:\3-2\AI Assitant Coding\Code Files>  & 'D:\3-2\AI Assitant Coding\Code Files\.venv\Scripts\python.exe' 'c:\U
tensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '65190' '--' 'D:\3-2\AI Assitant Co
py'
Alice's attendance: Present
Bob's attendance: Absent
Charlie's attendance: Not Recorded
PS D:\3-2\AI Assitant Coding\Code Files> & "D:\3-2\AI Assitant Coding\Code Files\.venv\Scripts\Activate.ps1"
(.venv) PS D:\3-2\AI Assitant Coding\Code Files> deactivate
PS D:\3-2\AI Assitant Coding\Code Files> python -m pydoc Task95
Help on module Task95:

NAME
    Task95

DESCRIPTION
    # Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
    # def reverse_string(s):
    #     """
    #     Reverses the given string.

CLASSES
    builtins.object
        Attendance

    class Attendance(builtins.object)
     |  # Generate 1.Create a module attendance.py with functions:mark_present(student),mark_absent(student),
     |  # get_attendance(student) 3.Generate and view documentation
     |  # attendance.py
```

Python: module Task95

D:/3-2/AI%20Assitant%20Coding/Code%20Files/Task95.html

index
**Task95** d:\3-2\ai assitant coding\code files\task95.py

# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
# def reverse_string(s):
# """
# Reverses the given string.

**Classes**

builtins.object
Attendance

class **Attendance**(builtins.object)
# Generate 1.Create a module attendance.py with functions:mark_present(student),mark_absent(student),
# get_attendance(student) 3.Generate and view documentation
# attendance.py

Methods defined here:

**__init__**(self)
Initializes the Attendance class with an empty attendance record.

**get_attendance**(self, student)
Retrieves the attendance status of a student.

Args:
student (str): The name of the student whose attendance status is to be retrieved.

Returns:
str: The attendance status of the student ("Present", "Absent", or "Not Recorded").

**mark_absent**(self, student)
Marks a student as absent.

Args:
student (str): The name of the student to be marked absent.

**mark_present**(self, student)
Marks a student as present.

New tab

Pydoc module Task95

localhost:2329/Task95.html

**Classes**

builtins.object
Attendance

class **Attendance**(builtins.object)
# Generate 1.Create a module attendance.py with functions:mark_present(student),mark_absent(student),
# get_attendance(student) 3.Generate and view documentation
# attendance.py

Methods defined here:

**__init__**(self)
Initializes the Attendance class with an empty attendance record.

**get_attendance**(self, student)
Retrieves the attendance status of a student.

Args:
student (str): The name of the student whose attendance status is to be retrieved.

Returns:
str: The attendance status of the student ("Present", "Absent", or "Not Recorded").

**mark_absent**(self, student)
Marks a student as absent.

Args:
student (str): The name of the student to be marked absent.

**mark_present**(self, student)
Marks a student as present.

Args:
student (str): The name of the student to be marked present.

Data descriptors defined here:

**__dict__**
dictionary for instance variables

**__weakref__**
list of weak references to the object

# Problem 5: File Handling Function

**Consider the function:**

**def read_file(filename):**

**with open(filename, 'r') as**

**f: return f.read()**

# Task:

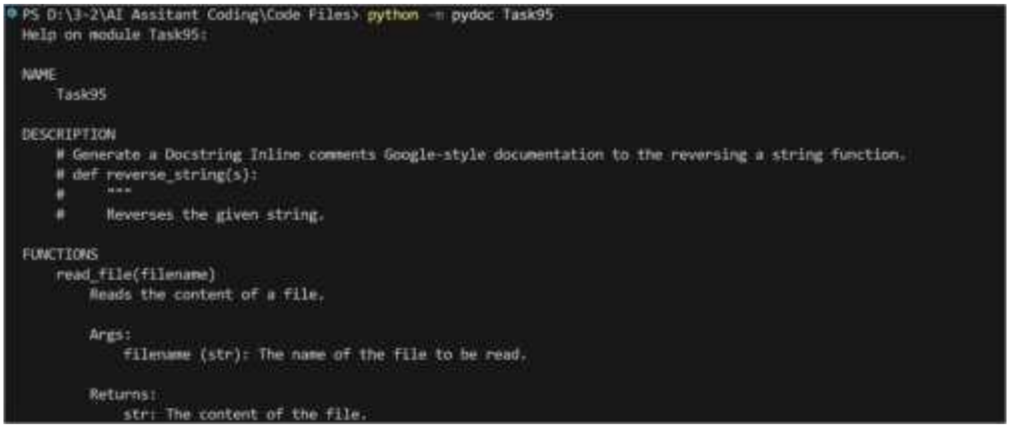
1. **Write documentation using all three formats.**
2. **Identify which style best explains exception handling.**
3. **Justify your recommendation.**

**Prompt: Generate a code with def read_file(filename):,with open(filename, 'r') as f:,return f.read(). Generate documentation using all three formats.2. Identify which style best explains exception handling.**

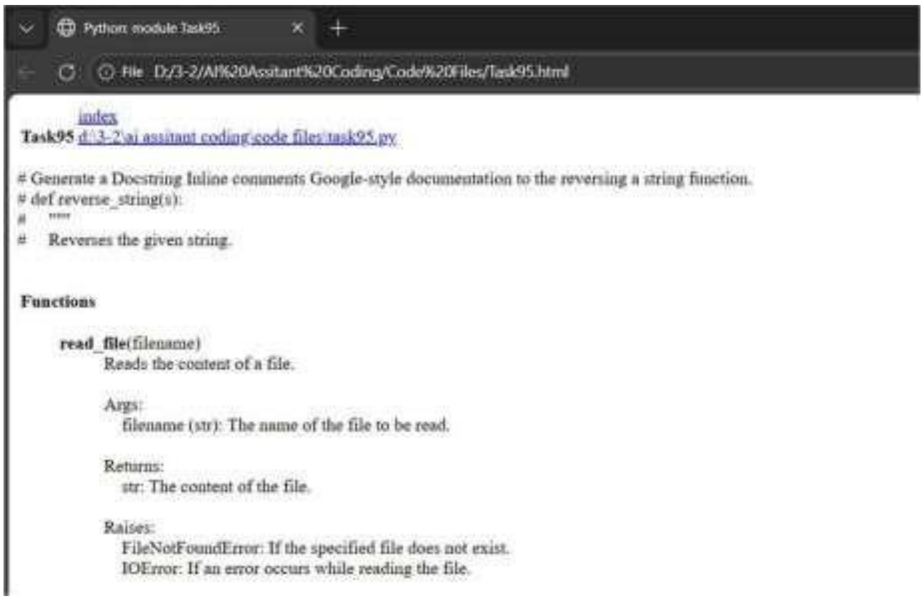
**Code:**

```python
def read_file(filename):
    """
    Reads the content of a file.

    Args:
        filename (str): The name of the file to be read.

    Returns:
        str: The content of the file.

    Raises:
        FileNotFoundError: If the specified file does not exist.
        IOError: If an error occurs while reading the file.
    """
    try:
        with open(filename, 'r') as f:
            return f.read()  # Return the content of the file
    except FileNotFoundError:
        return f"Error: The file '{filename}' was not found."
    except IOError as e:
        return f"Error: An I/O error occurred while reading the file. Details: {e}"
# Example usage
if __name__ == "__main__":
    filename = "example.txt"
    content = read_file(filename)
```

**Output:**



```
PS D:\3-2\AI Assitant Coding\Code Files> python -m pydoc Task95
Help on module Task95:

NAME
    Task95

DESCRIPTION
    # Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
    # def reverse_string(s):
    #     """
    #     Reverses the given string.

FUNCTIONS
    read_file(filename)
        Reads the content of a file.

        Args:
            filename (str): The name of the file to be read.

        Returns:
            str: The content of the file.
```





index
**Task95** d:\3-2\ai assitant coding\code files\task95.py

# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
# def reverse_string(s):
#     """
#     Reverses the given string.

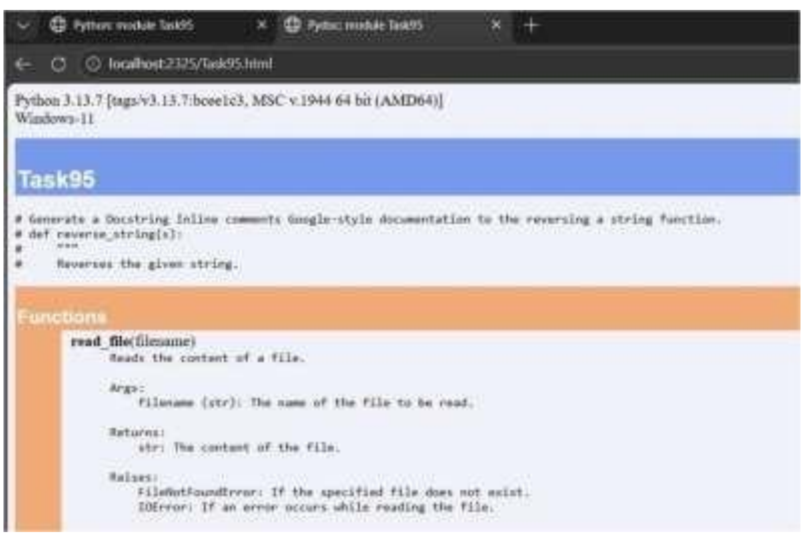
**Functions**

**read_file**(filename)
Reads the content of a file.

Args:
filename (str): The name of the file to be read.

Returns:
str: The content of the file.

Raises:
FileNotFoundError: If the specified file does not exist.
IOError: If an error occurs while reading the file.





Python 3.13.7 [tags/v3.13.7:bcee1c3, MSC v.1944 64 bit (AMD64)]
Windows-11

**Task95**

```
# Generate a Docstring Inline comments Google-style documentation to the reversing a string function.
# def reverse_string(s):
#     """
#     Reverses the given string.
```

**Functions**

**read_file**(filename)

```
Reads the content of a file.

Args:
    filename (str): The name of the file to be read.

Returns:
    str: The content of the file.

Raises:
    FileNotFoundError: If the specified file does not exist.
    IOError: If an error occurs while reading the file.
```